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PROJECT ARCHIVE

Spar Midrand 80 kWp

Client	Spar Group
Site	Spar Midrand
Location	Midrand, Gauteng
System	80 kWp + 100 kWh BESS
Contract	PPA
Project owner	Mathakgale
Status	Active — current stage: Quality Control
Start -> target handover	01 Sept 2026 -> 15 Dec 2026
Compiled	09 Nov 2026 by Mathakgale

Stage summary

Stage	Status	Completed	Tasks done
01 Site Visit	Done	10 Sept 2026	2 / 2
02 Design	Done	01 Oct 2026	—
03 BOQ	Not started		—
04 Procurement	Not started		—
05 Installation	Not started		—
06 Quality Control	In progress		0 / 1
07 As-Built	Not started		—
08 Handover	Not started		—

Contents

01 Site Visit

- Roof photo.jpg

02 Design

- Cable schedule Rev 00 (cable schedule)
- sse-dp037-rev-00.pdf (design pack)
- PV layout.dwg

03 BOQ

- BOQ Rev 00 (BOQ / costing)

04 Procurement

- Sungrow 125kW Grid Tied (Datasheet)

05 Installation

Checklist and notes only

06 Quality Control

- Spar Midrand – PV QC Checklist – 01 Nov 2026 (checklist / report)

07 As-Built

Checklist and notes only

08 Handover

Checklist and notes only

01 Site Visit

Status: Done (completed 10 Sept 2026)

Checklist

	Task	Assignee	Due	Done on
[x]	Take site photos	Thabo		05 Sept 2026
[x]	Shading assessment			

Notes

Site findings

Roof is IBR sheeting, good condition -> suitable for rail mounting. Main DB in plant room.

- Shading from water tank (east)
- Meter: Eastron SDM630

[x] Photos taken
[ ] Confirm roof load

Item	Value
Roof area	620 m²
Tilt	8°

Roof photo.jpg



02 Design

Status: Done (completed 01 Oct 2026)

Notes

Design approved by client on 28 Sep.

Cable schedule Rev 00

AC cable schedule

From	To	Ib (A)	Length (m)	Cable size	VD %	Breaker	Earth	Status
Inverter	MSB	120	60	95	0.73%	160	50	OK

DC cable schedule

From	To	Type	Current (A)	Length (m)	Cable size	VD %	Protection	Status
String 1	Inverter	PV String	14	45	6	0.74%	25	OK

DESIGN PACK – drawing sheet 1

DESIGN PACK – sheet 2

03 BOQ

Status: Not started

BOQ Rev 00

	SPAR MIDRAND 80 KWP QUOTATION	Project	Spar Midrand 80 kWp	Date	2026-10-05
Client	Spar Group	Revision	00		
Project Number	SSE-P-101	Prepared by	Mathakgale		
Document Number		Checked by			
Site / Address	Spar Midrand	System size	80 kWp · 100 kWh		

QUOTATION – PRICES IN ZAR

Item	Description	Unit	Qty	Rate (excl. VAT)	Amount (excl. VAT)
1	PV MODULES				
1.1	JA Solar 615 W bifacial	No.	130	R 2 312,50	R 300 625,00
	Subtotal – PV Modules	R 300 625,00			
2	INVERTERS				
2.1	Sungrow SG80CX	No.	1	R 89 544,00	R 89 544,00
	Subtotal – Inverters	R 89 544,00			
3	LABOUR & INSTALLATION				
3.1	Installation team	Days	12	R 11 875,00	R 142 500,00
	Subtotal – Labour & Installation	R 142 500,00			

Rate per Wp (excl. VAT)	R 6,99
Rate per kWp (excl. VAT)	R 6 991,28
Rate per kWh storage (excl. VAT)	R 5 593,02
Subtotal (excl. VAT)	R 532 669,00
Contingency (5%)	R 26 633,45
Total excl. VAT	R 559 302,45
VAT (15%)	R 83 895,37
TOTAL incl. VAT	R 643 197,82

Notes:

• All prices in South African Rand (ZAR). Quotation valid for 30 days from the date above.

• Quantities are based on the current design and may change after final site verification.

• Contingency covers unforeseen site conditions; unused contingency is not invoiced.

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04 Procurement

Status: Not started

SG125CX-P2

Multi-MPPT String Inverter for 1000 Vdc System



HIGH YIELD

- 12 MPPTs with max. efficiency 98.5%
- DC 15A current input, compatiable with over 500W+ PV module
- Dynamic shading optimization mode

SMART O&M

- Key component diagnosis and protection
- Smart IV Curve Diagnosis
- Grid fault record function, easy for remote O&M

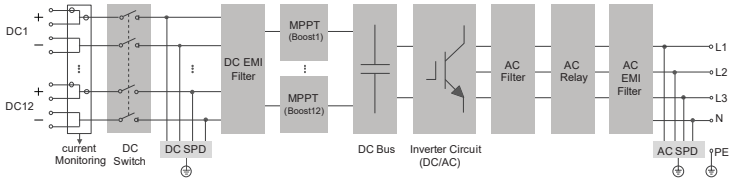
LOWER INVESTMENT

- Compatible max. 240mm² Al AC cables
- Drawer-style cable sealing plate support AC cable pre-assembly

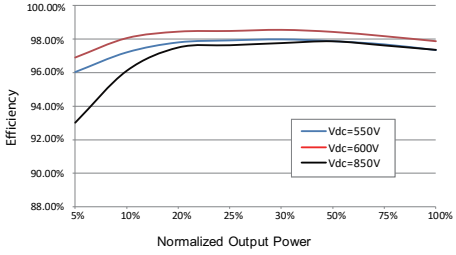
PROVEN SAFETY

- IP66 protection and C5 Anti-corrosion
- DC Type I+II SPD, AC Type II SPD
- Support AFCI 2.0 function

CIRCUIT DIAGRAM



EFFICIENCY CURVE



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Type designation	SG125CX-P2
Input (DC)	
Recommended max. PV input power	175 kW
Max. PV input voltage	1100 V
Min. PV input voltage / Startup input voltage	180 V / 200 V
Rated PV input voltage	600 V
MPP voltage range	180 – 1000 V
No. of independent MPP inputs	12
No. of PV strings per MPPT	2
Max. PV input current	360 A (30 A *12)
Max. DC short-circuit current	480 A (40 A * 12)
Max. current for DC connector	20A
Output (AC)	
Max. AC Output power	125 kVA (415 V @ 50 °C) *
Rated AC output apparent power	125 kVA (415 V @ 50 °C) *
Max. AC output current	181.1 A
Rated AC output current(at 230V)	181.1 A
Rated AC voltage	3 / N / PE, 230 / 400 V ; 3 / N / PE, 240 / 415 V
AC voltage range	320 – 480 V
Rated grid frequency	50 Hz / 60 Hz
Grid frequency range	45 – 55 Hz / 55 – 65 Hz
Harmonic (THD)	< 3 % (at rated power)
Power factor at rated power / Adjustable power factor	> 0.99 / 0.8 leading – 0.8 lagging
Feed-in phases / connection phases	3 / 3-N-PE
Efficiency	
Max. efficiency / European efficiency	98.5% / 98.3%
Protection	
Grid monitoring	Yes
DC reverse polarity protection	Yes
AC short circuit protection	Yes
Leakage current protection	Yes
Surge protection	DC Type I + II / AC Type II
Ground fault monitoring	Yes
DC switch	Yes
PV string monitoring	Yes
Q at night function	Yes
Arc fault circuit interrupter (AFCI)	Yes
PID recovery function	Yes
General Data	
Dimensions (W*H*D)	1020*795*360 mm
Mounting Method	Wall-mounting bracket
Weight	87 kg
Topology	Transformerless
Degree of protection	IP66
Corrosion	C5
Night power consumption	< 5 W
Operating ambient temperature range	-30 to 60 ℃
Allowable relative humidity range (non-condensing)	0 – 100 %
Cooling method	Smart forced air cooling
Max. operating altitude	4000 m (> 3000 m derating)
Display	LED, Bluetooth+APP
Optimizer	SP600S (Optional)
Communication	RS485 / Optional: WLAN, Ethernet
DC connection type	Evo2 (Max. 6 mm²)
AC connection type	OT / DT terminal (Max. 240 mm²)
Grid Compliance	IEC 62109-1, EN/IEC 61000-6-1/2/3/4, IEC 61727, IEC 62116, EN 50549-1/2, UTE C15-712-1, VDE V 0126-1-1, VDE-AR-N 4105:2018, VFR 2019, NC RfG, G99, UNE 217002, NTS, CEI 0-21 2019,CEI0-16 2019, NRS-097-2-1
Grid Support	Q at night function, LVRT, HVRT, active & reactive power control and power ramp rate control

* PV input voltage need over 630VDC

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05 Installation

Status: Not started

06 Quality Control

Status: In progress

Checklist

	Task	Assignee	Due	Done on
[ ]	Insulation resistance & polarity tests	Sipho	20 Nov 2026	

Spar Midrand – PV QC Checklist – 01 Nov 2026

PV QC Checklist · Spar Midrand · 01 Nov 2026 · Complete

PV QC Checklist

Check	Result	Comment
Module torque	Pass	All clamps 12 Nm
String Voc	Pass	738 V measured

Signed: S. Ndlovu

07 As-Built

Status: Not started

08 Handover

Status: Not started

Not included in this PDF

These files cannot be shown inside a PDF. They are in the ZIP archive in their original format.

Stage	File	Reason
Design	PV layout.dwg	.dwg cannot be shown in a PDF